

Joshua Ange

+1 (214) 500-3093 jwange@smu.edu joshange@gmail.com

EDUCATION

- Sep 2025 – Aug 2026 **University of Cambridge**
Churchill College
MPhil in Physics
- Aug 2021 – May 2025 **Southern Methodist University**
Dedman College of Humanities and Sciences
B.S. in Physics (with distinction), B.S. in Mathematics (Pure Math Concentration)
Minors in English and Computer Science
Summa Cum Laude, Honors in Liberal Arts
- Aug 2017 – May 2021 **North Central Texas College**
Associates of Science (General Studies)

RESEARCH EXPERIENCE

- Sep 25–Present **University of Cambridge**
Cavendish Astrophysics Group (B. Sherwin and B. Bolliet)
- May 24–Jul 24 **Stony Brook University**
Physics REU (N. Sehgal) — *Simulated high-resolution CMB maps to study and mitigate extragalactic foregrounds in preparation for CMB-HD [1]*
- May 24–May 25 **Southern Methodist University**
Darwin Deason Institute (M. Thornton) — *Studied modeling and optimization problems for high-dimensional and continuous-variable quantum computing [2, 3, 4]*
- Aug 22–May 25 Hamilton Research Fellowship (J. Meyers) — *Investigated CMB delensing on Λ CDM extensions and developed new tools to quantify lensing information of surveys [5, 6]*
- Jan 24–Aug 24 Bay Area Environmental Research Institute (M. Dainotti) — *Used ensemble machine learning techniques to classify high-redshift gamma-ray bursts [7]*
- May 23–May 24 Mayer Research Fellowship (E. Godat and R. Sudan) — *Investigated stylometric characteristics of physics literature with machine learning and literary techniques [8]*
- Mar 22–Aug 22 SuperCDMS SNOLAB (J. Cooley and R. Calkins) — *Studied influence of static charge on alpha particle measurements for dark matter background rejection [9]*

OTHER EXPERIENCE

- Jan 25–May 25 **Southern Methodist University**
Senior Editor (Journal of Undergraduate Research) — *Advised editor-in-chief and student research publication*
- Aug 24–Dec 24 Undergraduate Teaching Assistant (Darwin Deason Institute) — *Developed content and graded for introductory quantum computing course*
- Aug 23–Dec 24 Editor-in-Chief (Journal of Undergraduate Research) — *Oversaw editorial operations and content of peer-reviewed student journal*
- Jan 23–Dec 23 Undergraduate Teaching Assistant (Department of Physics) — *Taught in undergraduate labs and graded introductory physics exams*
- Nov 22–Aug 23 Assistant Editor (Journal of Undergraduate Research) — *Edited and guided student research through publication process*
- Oct 21–May 23 Student Worker (University Honors Program) — *Facilitated Honors events, programming, and Honors Student Advisory Council*

Apr 21–Aug 22 **NASA VIPS**
Student Mentor — *Mentored NASA HUNCH (High school students United with NASA to Create Hardware) Design & Prototype students*

PRESENTATIONS

Core Work in Cosmology and the CMB...

- "Recovering CMB Structure Through Delensing on the Path to Precision Cosmology"
 - *SMU 2025 Physics Speaker Series*
- "Computing the All-Orders Lensing Trispectrum of the CMB"
 - *Texas Section American Physical Society Fall 2024 Conference*
- "CMB-HD Foregrounds: Source Detection and Removal"
 - *SBU 2024 Summer Symposium*
- "Cosmic Microwave Background Delensing and The Hubble Tension"
 - *SMU 2023 Research and Innovation Week Undergraduate Poster Session*
 - *Texas Section American Physical Society Spring 2023 Conference*
 - *Texas Section American Physical Society Fall 2023 Conference*
 - *SMU 2023 Jane & Jack Hamilton Undergraduate Research Scholars Reception*

Investigations in Adjacent Fields...

- "Optimization and Realization of Boson Sampling for True Random Number Generation using the Xanadu X8"
 - *SPIE 2025 Defense + Commercial Sensing Conference*
- "Exploring Stylometric and Formal Patterns in the Scholarly Impact of Scientific Literature"
 - *SMU 2024 Research and Innovation Week Undergraduate Poster Session*
 - *SMU 2024 Research and Innovation Week Graduate Poster Session*

Early Work in Physics...

- "Characterization of XIA UltraLo-1800 Response to Measuring Charged Samples for Dark Matter Backgrounds"
 - *SMU 2022 Summer Research Intensive 3-Minute Thesis*
 - *SMU Fall 2022 Research Symposium*
 - *Texas Section American Physical Society Fall 2022 Conference*
- "Lunar Supply Pod Impact Simulation"
 - *2021 STEM Showcase*

HONORS & AWARDS

- 2025 Graduate Research Fellowship (NSF): Awarded to outstanding graduate students in STEM fields (provides three years of tuition support and a stipend over five years, totaling $\sim$ \$150,000)
- 2025-2026 Churchill Scholarship (Cambridge): Awarded to 16 recipients annually from the U.S. to study at the University of Cambridge on the basis of academic/research achievement and scientific impact (includes tuition, airfare, and a stipend, totaling in around \$70,000)
- 2025 Dedman College Outstanding Senior (SMU): Honor given to one senior student in Dedman College who exemplifies exceptional scholarship, leadership, and service to the university
- Phi Beta Kappa Honor Society: Inducted for outstanding academic achievement in the liberal arts and sciences
- 2024 Physics Robert S. Hyer Award (SMU): Nominated and awarded for student who performs "at the highest academic level" in physics
- Hamilton Undergraduate Research Fellowship (SMU): Fellowship granting funding for undergraduate research with Joel Meyers across 2022–2023, 2023 Summer, 2023–2024, 2024–2025 terms
- Robert Mayer Undergraduate Research Fellowship (SMU): Fellowship granting funding for undergraduate research with Rajani Sudan and Eric Godat across 2023–2024 term
- 2023 Barry Goldwater Scholarship: The "most prestigious national undergraduate scholarship" for excellence in natural sciences, engineering, or math research in the United States (totaling \$15,000)
- 2023 Physics Undergraduate Scholarship (SMU): Nominated and awarded for "continued commitment to the study of physics"
- 2022 Summer Research Fellow (SMU): Named for demonstrating "exceptional work ethic, curiosity, independence, and intellectual rigor" in research
- President's Scholarship (SMU): Highest university scholarship possible from academic merit (includes full tuition, room, and board, totaling in over \$80,000 per year)
- University Honors Program (SMU): Awarded as top 10% of entering class
- 2021 National Finalist (NASA HUNCH): Highest possible award, granted for physics simulation of lunar and Martian supply pod landings
- 2020 National Finalist (NASA HUNCH): Highest possible award, granted for math model of lunar supply pod landings

PUBLICATIONS (PUBLISHED)

- [2] Elena R. Henderson, Jessie M. Henderson, Joshua Ange, and Mitchell A. Thornton. “Programming quantum computers with large language models”. In: *Quantum Information Science, Sensing, and Computation XVII*. Vol. 13451. International Society for Optics and Photonics. SPIE, 2025, p. 1345104. DOI: [10.1117/12.3049666](https://doi.org/10.1117/12.3049666).
- [3] Joshua Ange, Mason Tuller, Jessie M. Henderson, Elena R. Henderson, Bradley A. Moores, Duncan L. MacFarlane, and Mitchell A. Thornton. “Modeling and Simulation of Multiple-Valued and Nonlinear Quantum Photonic Components”. In: June 2025. DOI: [10.1109/ISMLV64713.2025.00032](https://doi.org/10.1109/ISMLV64713.2025.00032).
- [4] Joshua Ange and Mitchell A. Thornton. “Optimization and realization of boson sampling for true random number generation using the Xanadu X8”. In: *Quantum Information Science, Sensing, and Computation XVII*. Vol. 13451. International Society for Optics and Photonics. SPIE, 2025, p. 1345106. DOI: [10.1117/12.3053219](https://doi.org/10.1117/12.3053219).
- [5] Joshua Ange and Joel Meyers. “Improving constraints on models addressing the Hubble tension with CMB delensing”. In: *JCAP* 10 (2023), p. 045. DOI: [10.1088/1475-7516/2023/10/045](https://doi.org/10.1088/1475-7516/2023/10/045). arXiv: [2307.01662](https://arxiv.org/abs/2307.01662) [[astro-ph.CO](https://arxiv.org/archive/astro-ph)].
- [7] Maria Giovanna Dainotti, Shubham Bhardwaj, Christopher Cook, Joshua Ange, Nishan Lamichhane, Malgorzata Bogdan, Monnie McGee, Pavel Nadolsky, Milind Sarkar, Agnieszka Pollo, and Shigehiro Nagataki. “GRB Redshift Classifier to Follow up High-redshift GRBs Using Supervised Machine Learning”. In: *The Astrophysical Journal Supplement Series* 277.1 (2025), p. 31. DOI: [10.3847/1538-4365/adafa9](https://doi.org/10.3847/1538-4365/adafa9). arXiv: [2408.08763](https://arxiv.org/abs/2408.08763) [[astro-ph.HE](https://arxiv.org/archive/astro-ph)].
- [9] Joshua Ange, Robert Calkins, and Andrew Posada. “Characterization of XIA UltraLo-1800 response to measuring charged samples”. In: *JINST* 18.01 (2023), P01027. DOI: [10.1088/1748-0221/18/01/P01027](https://doi.org/10.1088/1748-0221/18/01/P01027). arXiv: [2209.08002](https://arxiv.org/abs/2209.08002) [[physics.ins-det](https://arxiv.org/archive/physics)].

PUBLICATIONS (IN PREPARATION)

- [1] Amanda McInnis, Joshua Ange, Neelima Sehgal, Josh Kable, and Isabelle Fite. “CMB-HD Foregrounds: Ultra-High-Resolution Simulations and Foreground Removal”. In preparation. 2025.
- [6] Joshua Ange and Joel Meyers. “Computing the All-Orders Lensing Trispectrum of the CMB”. In preparation for *JCAP*. 2025.
- [8] Joshua Ange, Eric Godat, and Rajani Sudan. “Stylometric and Formal Patterns in the Scholarly Impact of Scientific Literature”. In preparation for *SMU JoUR*. 2026.